

7 C The 3 lines of action of the forces should coincide at a point.

8 D For each mass, Tension + Upthrust = Weight $\Rightarrow$ Tension = Weight - Upthrust
Since rod is horizontal, tension is the same for both P and Q
Upthrust = $V\rho g$ is greater for P since liquid X is denser.
So weight of P is greater to get the same tension as Q.

9 D Let T be the tension in the string